SARA OMAR MAHMOUD

.NET Backend Developer

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PROFESSIONAL SUMMARY

.NET Backend Developer with 5+ years building enterprise security, surveillance, and business systems: license plate recognition, visitor and building management, command & control, access monitoring, ticketing & tolling, and sales operations. Specialized in ASP.NET Core APIs, Clean Architecture, SQL Server, and MongoDB, with real-time SignalR and Milestone XProtect integrations. Shipped 10+ production systems end-to-end.

SKILLS

- **Backend:** C#, .NET, ASP.NET Core, ASP.NET Web API, RESTful APIs, Entity Framework, LINQ, ASP.NET MVC, Swagger/OpenAPI
- **Databases:** Microsoft SQL Server (Stored Procedures, SqlTableDependency), MongoDB (NoSQL)
- **Architecture & Practices:** OOP, SOLID Principles, Clean Architecture, Dependency Injection, Design Patterns (Singleton), API Design, Database Design, Agile, Code Reviews
- **API Security & Reliability:** JWT Authentication, Role-based Authorization, In-Memory Caching (IMemoryCache), FluentValidation, Serilog, BCrypt Password Hashing
- **Real-Time & Integrations:** SignalR, Milestone XProtect SDK (Smart Client Plugins, Live View, Playback), MailKit
- **Desktop & Cross-Platform:** WPF, Windows Forms, .NET MAUI
- **Reporting & Maps:** RDLC Reports, QuestPDF, EPPlus (Excel), LiveCharts, GMap.NET
- **Tools & Deployment:** Git, GitHub, Docker, Postman, Visual Studio, VS Code, Advanced Installer, Visual Studio Installer Projects

PROFESSIONAL EXPERIENCE

Freelance .NET Backend Developer May 2025 – Present

- Architected and delivered the **Mandoobi** backend end-to-end within a 4-person team: a retail and wholesale sales management platform covering sales representatives, supervisors, warehouses, vehicle inventory, invoices, quotations, money collection, and returns.
- Built an ASP.NET Core 9 Web API with 70+ controllers on MongoDB, structured with Clean Architecture and Dependency Injection.
- Secured the API with JWT authentication, role-based authorization, BCrypt password hashing, and FluentValidation, and added in-memory caching (IMemoryCache) and Serilog structured logging.
- Implemented real-time updates with a SignalR hub, PDF quotations with QuestPDF, and email notifications with MailKit.
- Created 20 bilingual (Arabic/English) RDLC reports, including daily sales, inventory, profit margin, product performance, customer lifetime value, and salesperson performance.
- Implemented business logic and API integration for the WPF admin application on top of provided UI designs, and packaged it as a Windows installer.

.NET Backend Developer | MTI Holding — Cairo, Egypt March 2021 – May 2025

- As part of the development team, designed and engineered the backend for 10+ enterprise security, monitoring, and operations systems, including building management (BMS), visitor management (VMS), command & control center (CCC), entrance control, license plate recognition (LPR), and ticketing & tolling.
- Developed REST APIs with ASP.NET Web API and ASP.NET Core (Swagger) on MongoDB and SQL Server, and designed SQL Server databases with Entity Framework and stored procedures, serving WPF desktop applications and a .NET MAUI tablet app.

- Engineered real-time features with SignalR, using the Singleton pattern for shared hub connections, and SQL Server change notifications (SqlTableDependency).
- Produced 20+ RDLC reports for the Ticketing & Tolling System (cashier shifts, refunds, subscriptions, auditing), plus LPR dashboards with charts and Excel export.
- Built Milestone XProtect Smart Client plugins using the Milestone SDK, such as license plate recognition (LPR) inside the VMS, embedded live video streaming and playback into the Ticketing & Tolling System, and added video bookmarking and retrieval.
- Developed GIS mapping tools with GMap.NET, and packaged and deployed 4 desktop applications (BMS, VMS, CCC, and entrance control) as Windows installers with Advanced Installer and Visual Studio Installer Projects.
- Translated complex business requirements into database and application designs, modernized legacy systems into a layered architecture, and participated in peer code reviews.
- Presented live demos of our systems at **Cairo ICT 2022**, including a working ticketing & tolling lane mock-up, and explained features and technologies to clients and visitors.

KEY PROJECTS (MTI)

Security & Facility Management Platform

- Integrated BMS, VMS, CCC, entrance control, and time-limited camera access request modules.
- *ASP.NET Web API, ASP.NET Core, MongoDB, SignalR, WPF, .NET MAUI, RDLC*

LPR — Milestone XProtect Plugin

- Plate and car make recognition, black and white lists, live passing-car events, and video playback.
- *C#, WPF, Milestone SDK, MongoDB, SignalR, LiveCharts, EPPlus*

Ticketing & Tolling System

- Ticketing, tolling, monthly and UHF subscriptions, cashier shifts, and embedded Milestone video.
- *C#, WinForms, SQL Server, Entity Framework, RDLC, Milestone SDK*

Company Management & Archive System

- Companies, agencies, contracts, products, policies, ratings, and maintenance notifications.
- *C#, WPF, SQL Server, Entity Framework, Excel export*

EDUCATION

Bachelor of Computer Science | Faculty of Computer & Information Sciences, Ain Shams University | 2015 – 2020

LANGUAGES

Arabic (Native) | English (Professional Working Proficiency)